

Why do you want to be a Community Catalyst?

Engineering undergraduate programs have trended from skill-based, applied approaches towards a more theoretical and scientific knowledge focus. The real world practitioner experiences and integrative, team-based development approaches were gradually lost and replaced with lectures and exams. On the other hand, inductive pedagogies like Active Collaborative Learning (ACL), Project Based Learning (PBL) and Entrepreneurial Minded Learning (EML) are more student-centered and begin with a real world problem or observation that is introduced to the students. These techniques can lead to deeper student learning when properly implemented.

I have participated in many KEEN conferences and workshops, implemented EML teaching modules across many BME courses and share these entrepreneurial-minded projects at workshops we have hosted, conferences, as well as on EngineeringUnleashed.com (EU.com). Currently, we are developing of a Wearable Technology Innovation Center (WTIC) that brings design, research, engineering and business experts together to work on exciting and highly disruptive products. An example of one of these projects is wearable sensors to train athletes with correct jump landing techniques that prevent knee injuries in sports. The WTIC is dedicated to creating value by developing wearable technology devices, which could impact young athletes, workers and elderly people's health and safety. These real world themes motivate the PBL modules, starting with the entrepreneurial mindset, then incorporating engineering design skills and finally communicating solutions.

One of the first such modules was a collaboration with Santa Clara University and Nike Inc. to get freshman in our Introduction to Biomedical Engineering course to identify opportunities for an in-shoe pressure sensor system. We have also collaborated with Western New England University for a module that emphasizes technical biomechanics concepts with EML by engaging LTU coaches to interact with students to build swing sensors to help players improve their batting performance. A final example is the development of a cross-disciplinary course focused on the EML mindset, "Wearable Technology Design Studio". This was part of the College of Engineering's goal was to create a, junior-level, discipline-specific course to be part of the four-year engineering design sequence focused on EML. It also engaged College of Design students who shared their perspectives on user-centered design and excellent prototyping skills.

Recently, I have been working on developing Simulation-Based Active Learning (SAL) modules which provide opportunities for repeated design-implement experiences as a more authentic context for learning fundamental science content, engineering design and analytical capabilities. The aim is to provide more opportunities for students to experience engineering and to develop students who are better at adapting to new trends, embrace creativity and leadership, and to understand impacts on society and business.

Simulation-based active learning (SAL) has been used in high-risk professions such as aviation, military and medical education. Benefits include; identification and prevention of errors, efficiency, practice with motor skills, assessment of skills, student exposure to specific applications and situations that might be rare or difficult to experience due to lack of resources.

As an example in Orthopedic Surgery, simulation allows residents to make mistakes, receive feedback, and improve performance without the pressure of the clinical setting and risk to human patients. SAL can be implemented with low-fidelity or high-fidelity representations using a variety of materials;

synthetic models, animal models, cadavers, computational models and virtual reality. We have collaborated with Kettering University and Nursing students at Lawrence Technological University (LTU) to implement a series of SAL modules in BME 5303 "Engineering Applications in Orthopedics" to build students' experience, entrepreneurial mindset and communication skills in engineering design projects to solve challenges in Orthopedics.

In addition to the cards published on EU.com, we have engaged faculty, students and industry representatives in a KEEN sponsored workshop called "The QS4EML Roadshow" at five member institutions. Faculty in this subnet had access to EML resources and developed related course modules for their programs. Many of these modules were shared between the universities and some of the collaborations resulted in conference papers or other publications. I also strongly advocate for EML with my Biomedical Engineering colleagues. In the Wearable Technology course we included six guest instructors from across all four Collages at LTU. The LTU Industrial Design programs also adopted that course so that their students and Biomedical Engineering students would experience working on cross-disciplinary teams.

I have been an active member of the LTU KEEN faculty since 2011 and greatly appreciate the opportunities and resources that KEEN provides new engineering faculty. I think the role of KEEN Community Catalyst would allow me to have increased partnerships and leadership to bring EML to more faculty and students to enhance engineering pedagogy and professional development.

Which area(s) of focus interest you the most? You may pick more than one.

Publishing new content

Coaching/mentoring members

Providing feedback on website functionality

Member interviews and focus groups

How Many Hours A Week Do You Think You Can Dedicate?

4 (I am on sabbatical F2023-S2024 so its possible to do more)